CSE461: Introduction to Robotics

Hardware Project Guidelines

Objective

Design and build a smart autonomous robot using a microcontroller (Arduino or Raspberry Pi), multiple sensors, and actuators. The project should address a **real-world problem** and align with course objectives. Each idea must be approved by the lab faculty.

Required Components

- Microcontroller (Arduino or Raspberry Pi)
- Chassis or robot frame
- Power supply (battery or external)
- At least **3 types of sensors**
- At least **2 types of actuators**
- Jumper wires, breadboard

Project Criteria

- Must be **autonomous** and **problem-solving**.
- Simple **line-following or remote-controlled** robots are **not acceptable** as standalone projects. These may only be included as features within a larger, autonomous system.

Steps and Deliverables

1. Project Proposal

Submit **three project ideas**. Find the details in the given link :

CSE461 Project Proposal Template

2. Project Demonstration and Report Submission

- Prepare a presentation covering design, implementation, and challenges.
- Demonstrate the robot's autonomous functionality.
- Submit the **final project report** covering both functionality and sustainability aspects of the project:

Notes

- Properly cite all components and references.
- Plagiarism will result in penalties.

- Consult your instructor if you face technical difficulties.